AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 24 · WORKBOOK

Privacy by Choosing Less

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected minimal-data design.

Redesign the workshop form, challenge anonymity claims, share limited totals and complete the independent library plan.

Keep one habit: Keep purpose, necessary fields, audience and retention together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Fair for Whom?

Good Data, Bad Data

Confidence Is Not Correctness

W1 · Minimise the workshop-vote form

> Remove unnecessary fields before collection rather than promising to ignore them.

Your evidence record

- The fictional workshop vote asks for name, full date of birth, address, photograph and topic choice. Its only **purpose** is to count votes for drawing, music and coding. State which field to keep, which to remove and why.

  Explain why collecting the extra fields first is different from never collecting them.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Challenge the anonymous-record claim

> Explain how a key or photograph can preserve identifiability.

Your evidence record

- A teacher replaces names with codes and keeps a separate code-to-name key. A pupil calls the records anonymous. Explain the problem.

  Then explain why removing names while leaving photographs would not settle the question either.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Share the needed result with less exposure

> Check the counts without publishing individual slips.

Your evidence record

- The fictional class has drawing 8, music 7 and coding 5. Write the result needed to choose a workshop. A pupil suggests posting every vote slip online as proof. Explain a lower-exposure way to check and share the result.

  Why can a total about a one-person club still reveal a choice?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Design a different minimal-data task

> Keep one first attempt before hints or feedback.

Read and explain independently

A fictional library wants only the total number of books needing repair in each category: cover, page or binding. It does not need to contact borrowers.

A draft repair sheet asks for borrower name, home address, book repair category and borrower photograph.

Design a plan: state the **purpose**, required and removed fields, who handles source sheets, what totals are shared with the librarian, and who disposes of sheets at what agreed point. Use no real borrower details.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Design a different minimal-data task

W5 · Review an upload after your library plan

> Use smaller fictional inputs or paper arithmetic and explain a first response to accidental sharing.

Your evidence record

- Someone plans to upload the library repair sheet with borrower details to an unapproved AI tool to add three totals. Explain a smaller-**input** alternative and a no-upload method.

  If the sheet was already shared, state a suitable first response and one claim you must not make about deleting it.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Cross out names, faces, and ownership details unless an explicit need survives review; set a deletion rule for sorter records.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Challenge a claim of anonymity for **Privacy** and **data minimisation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Privacy** and **data minimisation**. Proposed explanation: “More data always improves a task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A book recommender requests genre, maximum pages, home address and a selfie. Keep only fields needed for genre and length matching. Explain one consequence of storing extra fields.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A genre-based book helper asks for genre, home address and a photograph.

Which fields are necessary?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A book helper needs reading topic and approximate difficulty. Someone asks it to collect home address too. Which information can be left out, and why?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 24 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A book recommender requests genre, maximum pages, home address and a selfie. Keep only fields needed for genre and length matching. Explain one consequence of storing extra fields.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fictional library wants only the total number of books needing repair in each category: cover, page or binding. It does not need to contact borrowers.

A draft repair sheet asks for borrower name, home address, book repair category and borrower photograph.

Design a plan: state the **purpose**, required and removed fields, who handles source sheets, what totals are shared with the librarian, and who disposes of sheets at what agreed point. Use no real borrower details.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

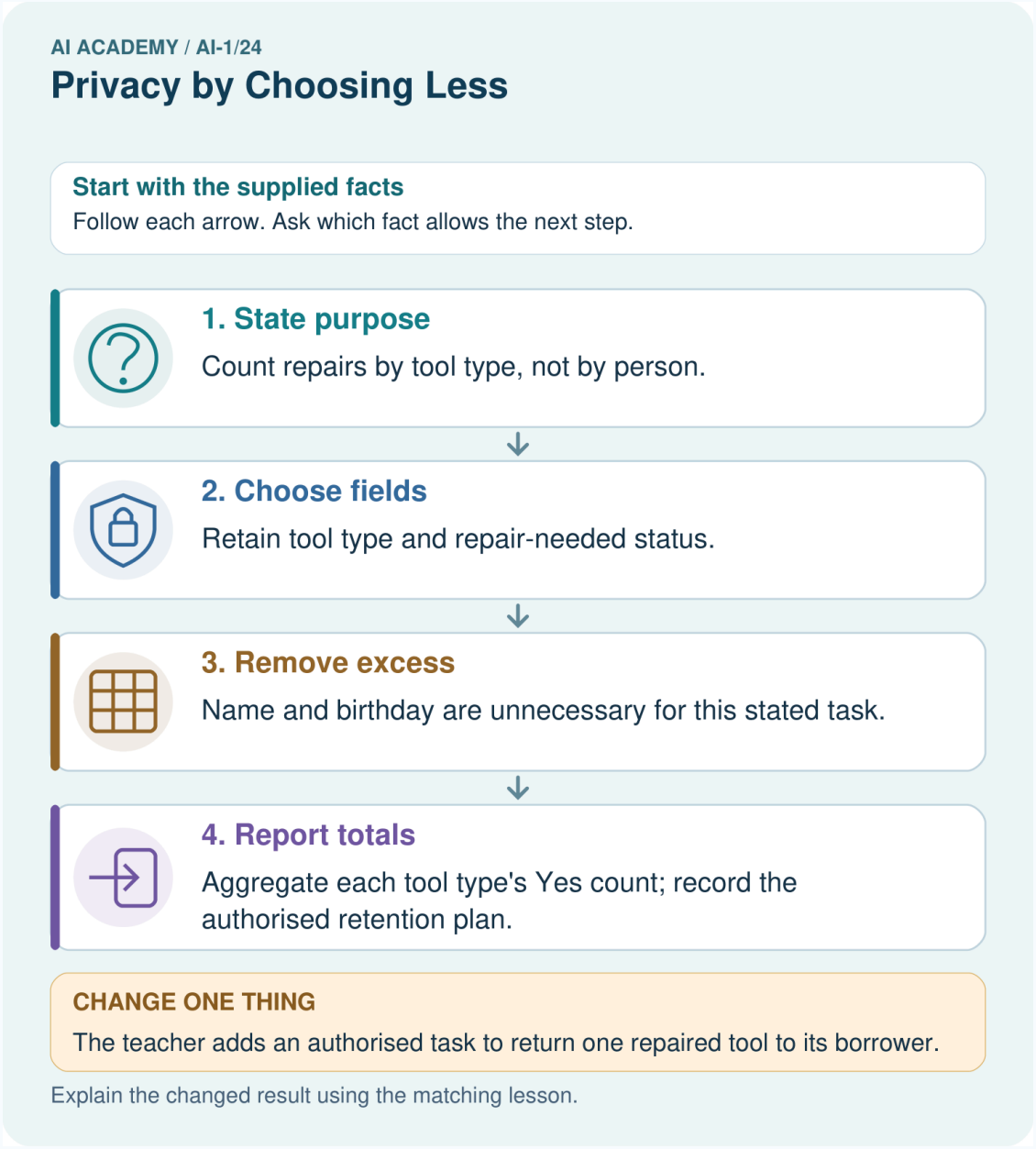
> Keep the explanation and the support you used.

Use this when helpful

A fictional teacher wants to count broken craft tools by type, without contacting borrowers.

Needed **output**: totals for scissors, ruler and hole-punch repairs.

Draft form requests pupil name, birthday, tool type and repair-needed Yes/No. No borrower follow-up is required.



Follow the labelled picture with the matching explanation. Open the vector diagram

For the original task, does adding an anonymous-looking pupil code improve the repair count?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Write the sorter’s **purpose**, necessary fields, access rules and deletion point before collecting any real classroom information.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Write the sorter’s **purpose**, necessary fields, access rules and deletion point before collecting any real classroom information.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional teacher wants to count broken craft tools by type, without contacting borrowers.
- Which labelled step matches this detail? Aggregate each tool type's Yes count; record the authorised **retention** plan.
- 1. State **purpose**
- 2. Choose fields
- 3. Remove excess
- 4. Report totals
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report totals Aggregate each tool type's Yes count; record the authorised **retention** plan.

A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals.

Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

Changed-case question

- Changed-case practice: The teacher adds an authorised task to return one repaired tool to its borrower. Which conclusion is supported by the supplied facts?
- 1. A minimal contact **mechanism** may now be needed for that tool.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: A minimal contact **mechanism** may now be needed for that tool. The **purpose** changed; reassess necessity rather than restoring every old field.

Repair and save your work

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, unit and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Review Confidence Is Not Correctness (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

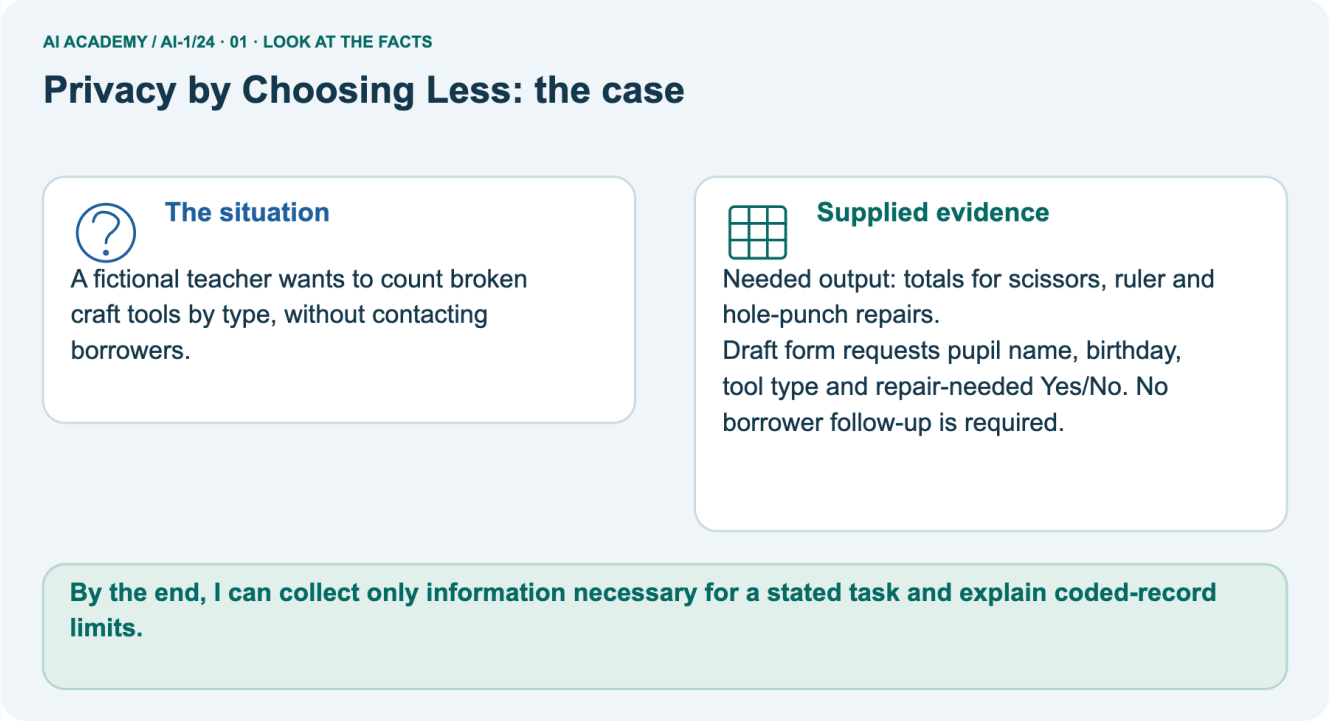
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Privacy by Choosing Less: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

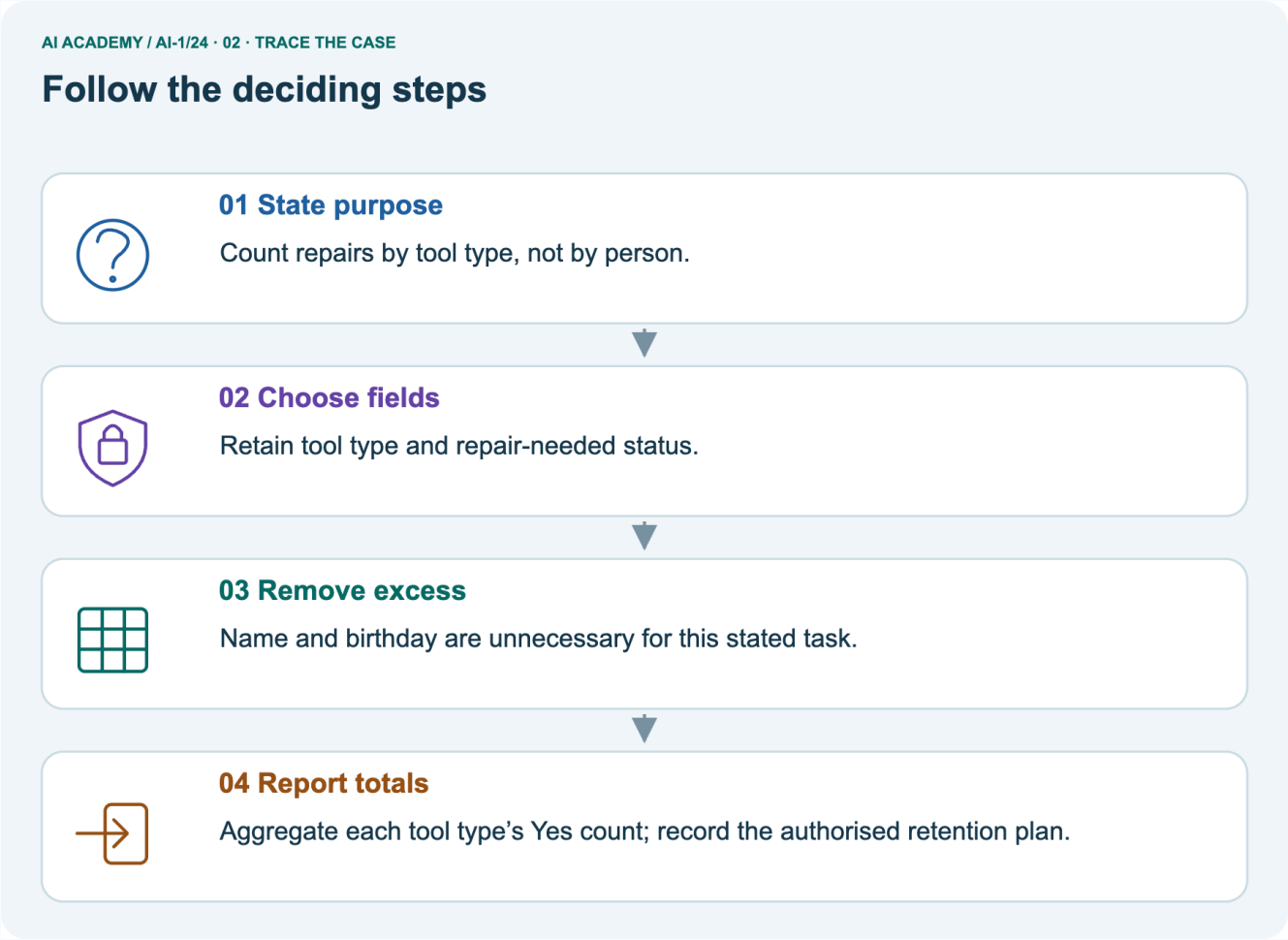
A fictional teacher wants to count broken craft tools by type, without contacting borrowers. Supplied facts: Needed **output**: totals for scissors, ruler and hole-punch repairs.

Draft form requests pupil name, birthday, tool type and repair-needed Yes/No. No borrower follow-up is required.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

State **purpose**: Count repairs by tool type, not by person. → Choose fields: Retain tool type and repair-needed status. → Remove excess: Name and birthday are unnecessary for this stated task. → Report totals: Aggregate each tool type's Yes count; record the authorised **retention** plan.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A small information plan for a fictional vote

AI ACADEMY / AI-1/24 · 03 · INSPECT THE LESSON MODEL

A small information plan for a fictional vote

Stage	Needed action	Privacy check
Collect	One topic per slip	No name or address
Count	Teacher totals choices	Keep slips restricted
Share	Three topic totals	No individual choices
Delete	After totals checked	Follow agreed plan

Purpose: choose a workshop topic. Collect the minimum, limit access, share the required result and agree how long the source slips are kept.

Follow the labelled picture with the matching explanation. Open the vector diagram

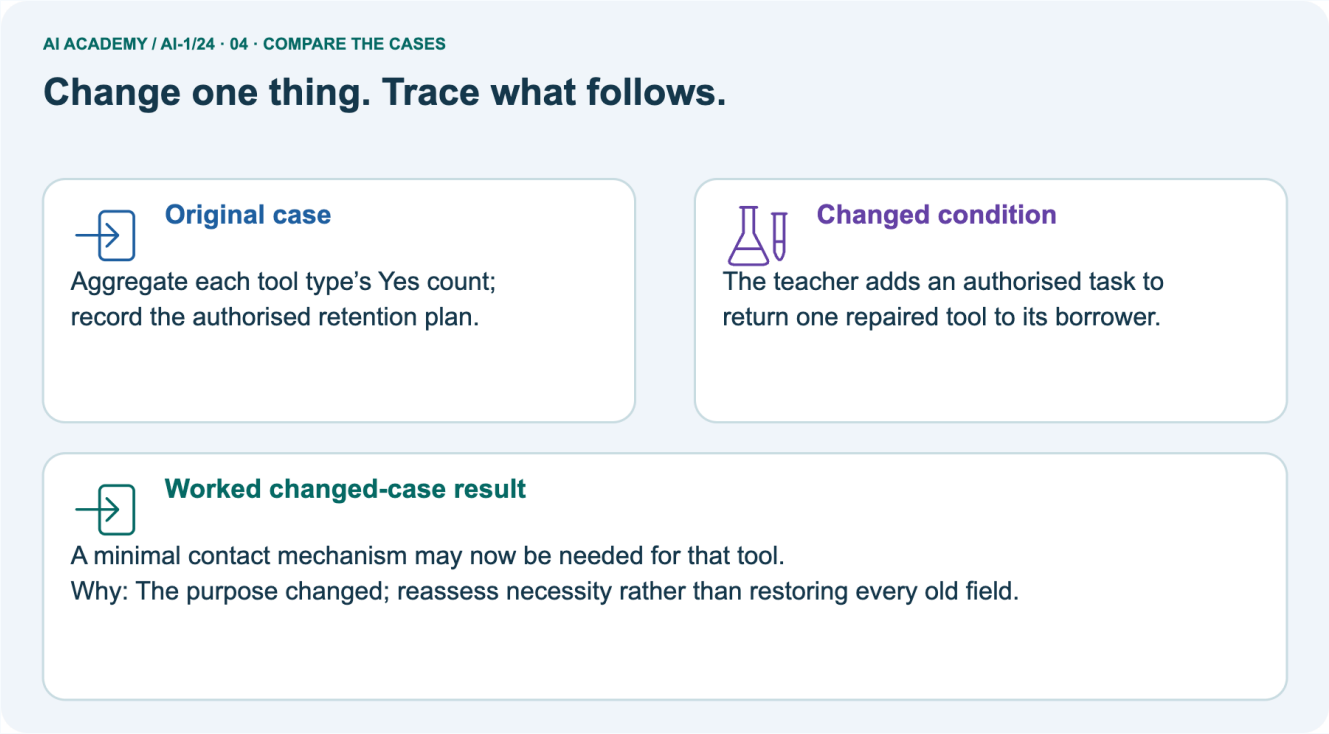
A small information plan for a fictional vote. Collect | One topic per slip | No name or address. Count | Teacher totals choices | Keep slips restricted. Share | Three topic totals | No individual choices.

Delete | After totals checked | Follow agreed plan

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

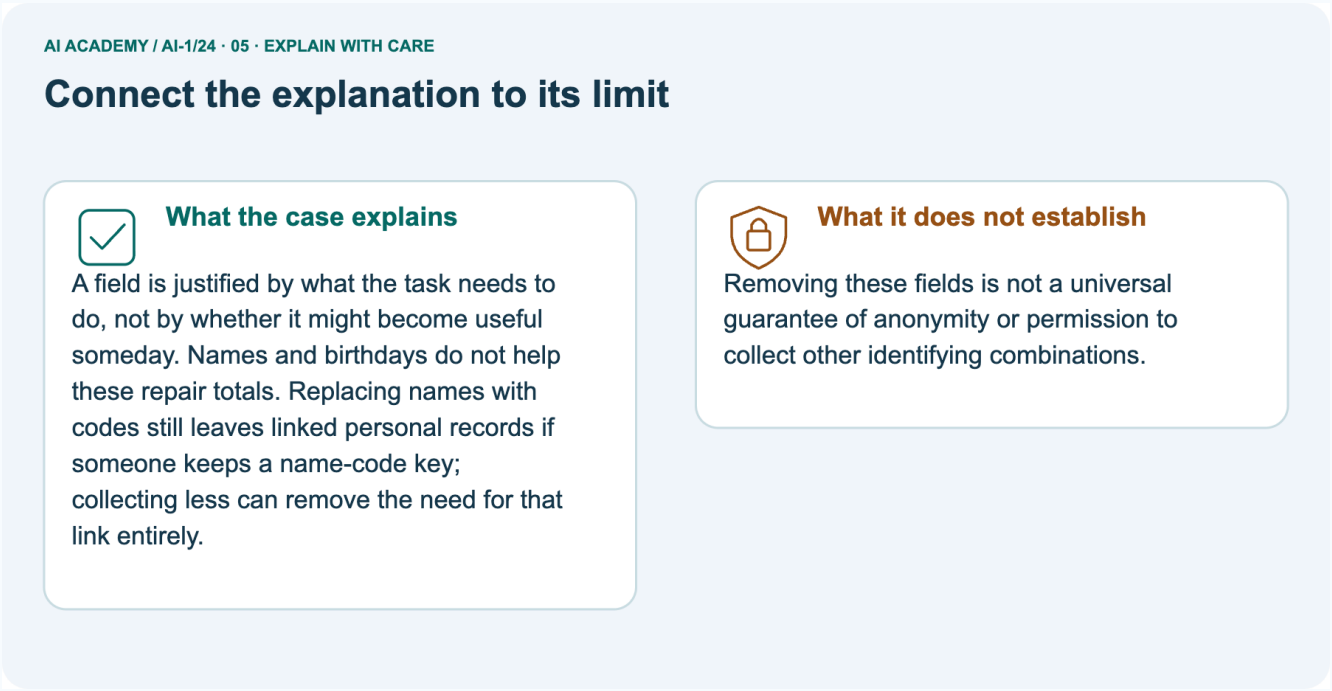
Original case: Aggregate each tool type's Yes count; record the authorised **retention** plan. Changed condition: The teacher adds an authorised task to return one repaired tool to its borrower.

Expected result: A minimal contact **mechanism** may now be needed for that tool. Reason: The **purpose** changed; reassess necessity rather than restoring every old field.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals.

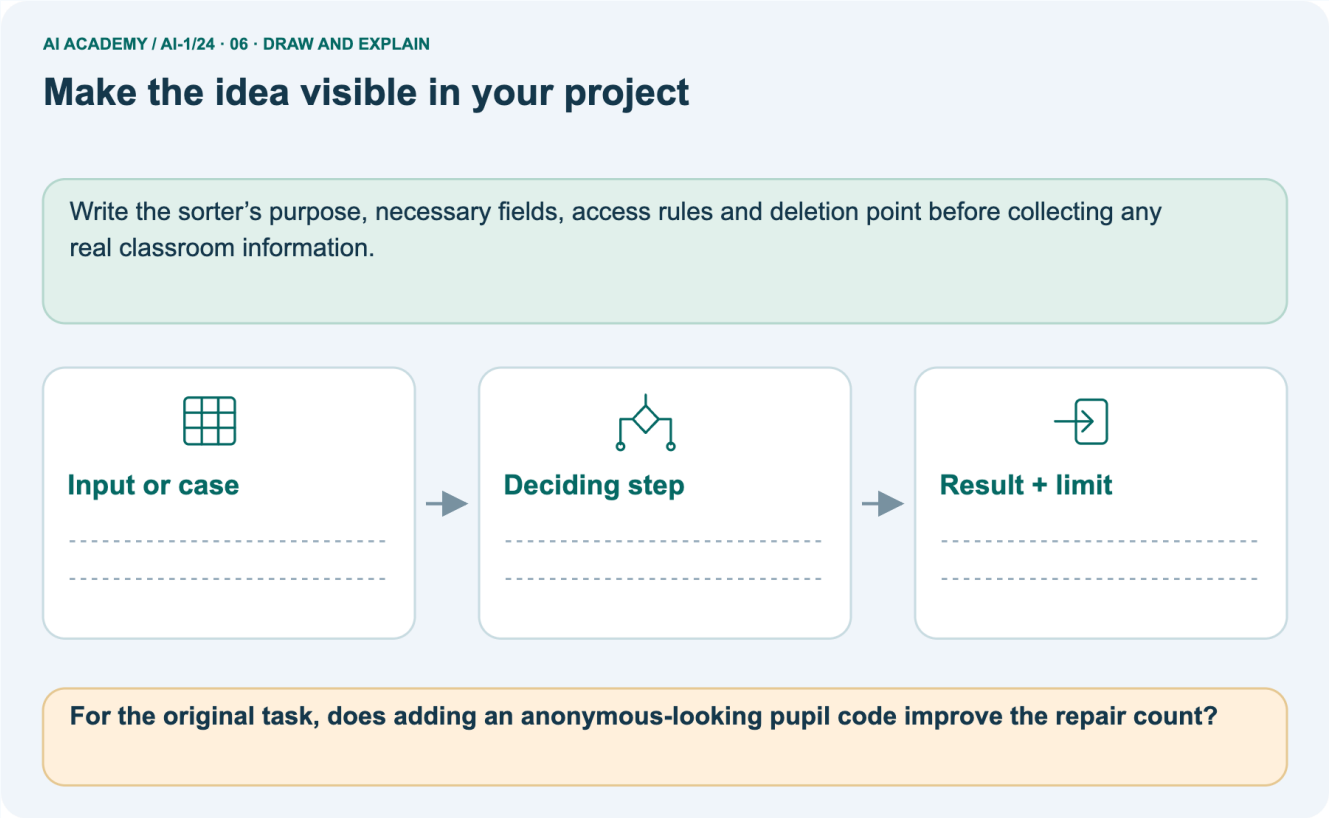
Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

Limit: Removing these fields is not a universal guarantee of anonymity or permission to collect other identifying combinations.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write the sorter's **purpose**, necessary fields, access rules and deletion point before collecting any real classroom information. Draw your **input**, deciding step, result and limit.

Supported question: For the original task, does adding an anonymous-looking pupil code improve the repair count?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Privacy by Choosing Less: the case

02. Follow the deciding steps

03. A small information plan for a fictional vote

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.